
Toward Mechanical Scientific Models: Fractal Custody and Governed Agent Experiments

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Abstract

1 The *Mechanical Scientific Method* treats scientific computation as a sequence of
2 governed state transitions in which evidence, transforms, and bounded claims re-
3 main inspectable rather than collapsed into model prose. We propose the term
4 *Mechanical Scientific Model* for a computational model class that satisfies explicit
5 requirements: explicit evidence state; governed and versioned state transitions;
6 separation of deterministic transforms from probabilistic model outputs; preserva-
7 tion of positive, null, negative, failed, blocked, timeout, abstention, and contradic-
8 tory outcomes; explicit claim ceilings; forward and reverse evidence traceability;
9 and successor-state recording rather than silent history rewriting. *Fractal Custody*
10 *Objects* (FCOs) and a *Fractal Custody Graph* (FCG) provide the custody
11 substrate for binding those transitions. HydraDG is the primary experimentally
12 evaluated implementation examined here: it preregisters hypotheses, freezes case
13 manifests and scorers, records raw prompt and response hashes, and preserves
14 terminal states without silent substitution. We report two preregistered HydraDG
15 studies (EXP-008, EXP-009) comparing flat prose versus structured FCG retrieval
16 for failure-prevention endpoints; both terminate as *underpowered*, so confirmatory
17 treatment-effect claims are not supported while null, negative, and failure evidence
18 remains in custody. We do not claim that Mechanical Scientific Models are an es-
19 tablished external field term; we offer them as a proposed model class and report
20 bounded custody behavior from one implementation.

21 1 Introduction

22 Large language model agents are evaluated on task success, benchmark scores, and human prefer-
23 ence [2, 3]. Less often preserved are the conditions under which experiments fail: swapped models,
24 stale runtime selection, partial matrix execution, or JSON outputs that never parse. When negative
25 and null cells disappear from the public record, later work cannot distinguish *no effect* from *no*
26 *measurement*.

27 This paper separates five layers that are often conflated in agent-evaluation reporting:

- 28 • **Method** — the *Mechanical Scientific Method*: governed transitions among explicit evi-
29 dence states rather than narrative collapse of runs into conclusions.
- 30 • **Model class** — *Mechanical Scientific Models* (proposed here; not an established external
31 terminology): computational systems whose state transitions are mechanically exposable
32 and failure-complete.
- 33 • **Custody substrate** — FCO/FCG receipts that bind source bytes, transforms, derived evi-
34 dence, bounded claims, and successor state.

- **Primary implementation** — HydraDG, the experimentally evaluated implementation whose preregistered EXP-008 and EXP-009 results are reported here.
- **Systems validation and companion exemplars** — HydraLamp perturbation and replay tests evaluate custody mechanics only; Vithia is a bounded companion implementation carried at zero primary evidentiary weight for EXP-008/009.

HydraDG instantiates the proposed model class through a custody-first experimental contract. Every actor—human operator, frontier agent, local model runtime, or deterministic script—must emit or be represented by a handoff receipt before its output is promoted. Receipts bind actor identity, host, repository state, input hashes, raw output hashes, evidence class, claim ceiling, and signature state (hashes are identity commitments, not digital signatures unless an authorized signing operation occurred). Probabilistic model outputs are never treated as authority; they are evidence objects subject to deterministic scoring and graph append rules.

This paper reports terminal results from the IC Failure Learning structured-retrieval falsification lane executed in HydraDG. We do *not* claim that structured FCG context improves failure prevention; preregistered EXP-008 and EXP-009 closed as *underpowered* with *effect not established*. We also do not treat exploratory secondary patterns as promoted findings, and we do not promote Vithia to primary evidence for those experiments.

2 Related Work

Agent evaluation. Benchmark suites measure capability but rarely require publication of failed runs, malformed generations, or infrastructure blocks [2, 3]. HydraDG complements benchmarks with mandatory negative capture at the custody layer.

Retrieval and structured context. Retrieval-augmented generation and graph-augmented memory systems assume structured context helps [1, 4]. Falsification requires frozen contracts, explicit null retention, and power-aware verdicts; HydraDG enforces both.

Provenance, integrity, and reproducibility. W3C PROV-O and workflow provenance systems represent process and research-object lineage [12, 13], while append-only Merkle audit logs provide established tamper-evident integrity mechanisms [11]; HydraDG does not claim to invent these primitives. Recent work surveys evidence tracing and execution provenance for LLM agents [15], and separate work argues for preserving negative ML results rather than selective omission [14]. Preregistration, FAIR principles, and nanopublication-style lineage further support reproducibility [5–7]. HydraDG integrates rather than replaces these foundations: probabilistic outputs, deterministic transforms, malformed cells, failures, abstentions, blocked states, and underpowered terminations remain typed evidence with explicit claim-promotion ceilings enforced via FCO/FCG handoff receipts. Companion custody-framework preprints (deferred to camera-ready de-anonymization) define FCO/FCG formally; they establish framework provenance, not the HydraDG empirical results reported here.

3 Framework

3.1 Mechanical Scientific Models

We propose the term *Mechanical Scientific Model* for a computational scientific system whose internal evidence-state transitions are mechanically exposable rather than inferred post hoc from model prose. We do not claim that this term is already established in the literature; it names a proposed model class whose required properties can be checked against an implementation.

Let S_t denote the governed evidence-and-claim state at step t . A Mechanical Scientific Model advances by governed transforms

$$S_t \xrightarrow[E_t]{T_t} S_{t+1},$$

where T_t is a versioned, auditable transform (deterministic scorer, ingest pass, governed model call, or tool invocation) and E_t is the evidence bound to that transition. Every transition must emit or bind:

F1: Proposed hierarchy (conceptual; not treatment-effect evidence)

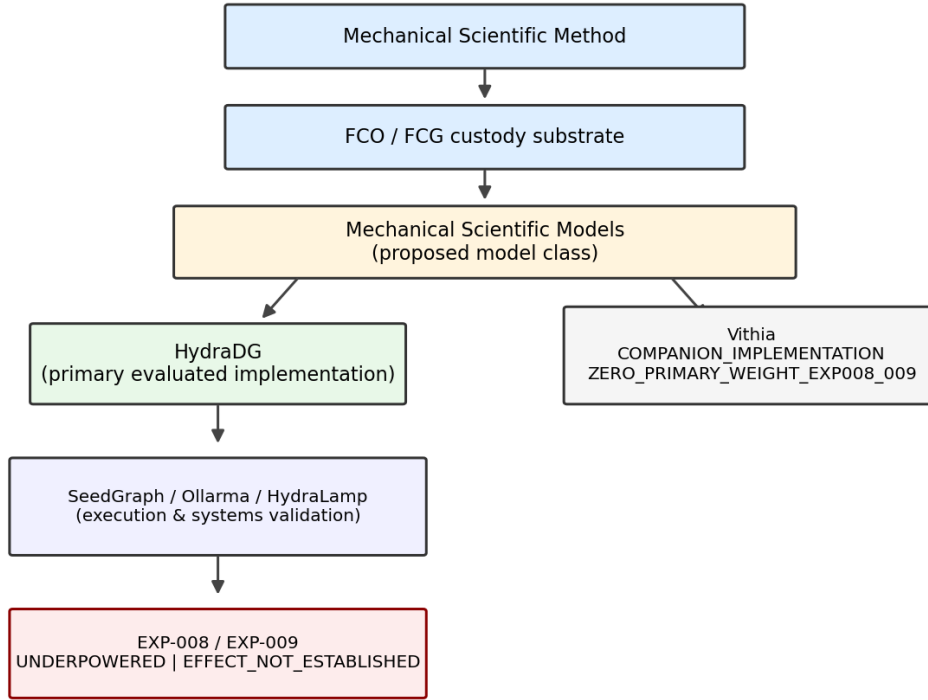


Figure 1: Proposed intellectual hierarchy (conceptual; not empirical effect evidence). HydraDG is the primary experimentally evaluated implementation; Vithia is a companion implementation with zero primary weight for EXP-008/009; EXP-008 and EXP-009 remain *underpowered* with effect not established.

- 82 1. source or upstream evidence;
- 83 2. transform, tool, or model identity (with deterministic vs. probabilistic output separation);
- 84 3. derived evidence or parser state;
- 85 4. a bounded claim with an explicit ceiling; and
- 86 5. an artifact or state delta recorded as successor state rather than silent history rewriting.

87 Required properties include: explicit evidence state; governed and versioned state transitions; deter-
 88 ministic vs. probabilistic output separation; preservation of positive, null, negative, failed, blocked,
 89 timeout, abstention, and contradictory outcomes; explicit claim ceilings; forward and reverse ev-
 90 idence traceability; and successor-state recording. Figure 1 situates the Mechanical Scientific
 91 Method, FCO/FCG custody, the proposed model class, HydraDG as the evaluated implementation,
 92 Vithia as a companion exemplar, and execution components used in this submission.

93 3.2 Custody objects

94 An **FCO** (*Fractal Custody Object*) materializes one bounded transformation: e.g., a model call,
 95 scorer pass, or ingest operation. An **FCG** (*Fractal Custody Graph*) append records directed edges
 96 between FCOs with hash-linked roots [11]. FCO/FCG provide the custody substrate on which
 97 Mechanical Scientific Models record transitions; HydraDG is one implementation that preserves
 98 failures, nulls, and malformed outputs in custody (a failure-complete *behavior*, not a redefinition of

99 FCO/FCG). Claim ceilings (exploratory mechanistic falsification, descriptive only, etc.) are declared
100 at preregistration and enforced at promotion time.

101 **3.3 Experimental freeze**

102 Each experiment declares:

- 103 • frozen case manifest and prompt-contract hashes;
- 104 • conditions (e.g., flat prose vs. structured FCG);
- 105 • models with expected runtime digests;
- 106 • primary endpoint and aggregation rule;
- 107 • explicit gates for confirmatory vs. exploratory claims.

108 Execution emits START_RECEIPT, per-cell raw outputs with prompt/response SHA-256, parser state,
109 and terminal receipts. Integrity failure (duplicate keys, missing dependencies, silent model substi-
110 tution) blocks promotion; scientific failure (timeout, abstention, malformed JSON) is retained as
111 terminal evidence.

112 **3.4 Governed local execution**

113 Where preregistered, local model calls route through a governed bridge that records selection age,
114 swap posture, and resolved digest. Direct runtime invocation and governed invocation are separate
115 evidence classes; one is not silently labeled as the other.

116 **4 Experimental Setup**

117 **4.1 Task family**

118 The IC Failure Learning suite uses synthetic agent-failure vignettes with preregistered endpoints
119 such as E06 (prevents exposure of out-of-vault media). Cases are drawn from a frozen CASES.jsonl
120 manifest; each case receives multiple replicates per condition.

121 **4.2 Conditions and models**

122 **C0 (FLAT_PROSE):** failure-handling guidance as unstructured text.

123 **C1 (STRUCTURED_FCG):** the same semantic content composed through a frozen FCG retrieval
124 contract.

125 EXP-008 and EXP-009 used local models qwen3:1.7b and qwen2.5-coder:7b under identical
126 case and scorer contracts. Thinking/reasoning modes were held consistent with preregistration. Pri-
127 mary aggregation: majority of three replicates per case.

128 **4.3 AI and agent methodology disclosure**

129 Frontier coding agents assisted with repository tooling and manuscript preparation; they are not
130 listed as authors. All scientific endpoints come from preregistered local model executions with
131 hashed prompts and responses. Routine editing does not constitute a reported experimental interven-
132 tion.

133 **4.4 Power and claim gates**

134 E06 endpoints are known limited-power under the frozen case set. Confirmatory ordering claims
135 require preregistered discordant pairs; exploratory secondary statistics are quarantined from primary
136 promotion.

Table 1: Terminal preregistered studies (HydraDG IC Failure Learning). *Underpowered* denotes insufficient paired evidence for confirmatory promotion, not proof of null effect.

Study	Primary verdict	Raw cells	Valid parse rate
EXP-008	UNDERPOWERED	300	0.907
EXP-009	UNDERPOWERED	300	0.883

Table 2: Systems-validation outcomes (custody mechanics only; not treatment-effect evidence).

Validation	Scope	Outcome
Perturbation matrix	100 cells	100/100 chain verification
Synthetic tamper suite	8 modes	8/8 detected
Concurrent execution	10 runs	PASS (10 unique run IDs)
Replay/restart recovery	44 events	PASS
Live provider ladder	R0–R6	Bounded external failure preserved

5 Results

Table 1 summarizes terminal verdicts for preregistered studies. Both experiments executed the full raw matrix ($n = 300$ cells each) with complete custody append.

EXP-008. Structured FCG retrieval did not yield promotable confirmatory evidence on E06 under the frozen design ($n = 2$ paired cases per model stratum at aggregation scope). Non-zero malformed and abstention rates are preserved in custody.

EXP-009. Primary verdict remains UNDERPOWERED for confirmatory E06 ordering. An exploratory secondary pattern was recorded as directionally positive at the descriptive layer only; it is **not promoted** to a causal claim.

Stage-2 baseline. Prior IC Failure Learning stage-2 execution (414 scored rows across two models) established FAILURE_LEARNING_BEHAVIOR_IMPROVEMENT_NOT_ESTABLISHED, providing historical context for the structured-retrieval falsification lane.

5.1 Failure-preserving systems validation

The preregistered retrieval experiments above test whether structured context improves a frozen endpoint. A separate systems-validation suite asks a different question: does the custody mechanism preserve integrity under perturbation, tampering, concurrency, replay, and external execution failure? Table 2 summarizes source-verified outcomes from a frozen HydraLamp perturbation matrix; these results evaluate custody mechanics and do **not** constitute evidence that structured FCG context improves the EXP-008/009 endpoint.

The tamper suite used synthetic cases only (not real security incidents). In a live provider repair ladder, earlier infrastructure gates passed before a structured call hit an external quota limit; the failure remained a terminal auditable state rather than being erased or substituted. The same custody primitives have been instantiated in sibling systems for local-model orchestration, deterministic scientific-document atomization, offline anomaly detection, bounded bioinformatics, substrate generation, and operator-facing scientific workflow monitoring; these implementations motivate transfer studies but do not provide additional evidence for the treatment effect reported here.

6 Discussion

The principal contribution of this submission is not evidence that structured FCG retrieval improves failure prevention. EXP-008 and EXP-009 remain *underpowered*; the empirical retrieval effect on the preregistered E06 endpoint is *not established*. The broader methodological proposition is that scientific computational models can expose their evidence-state transitions mechanically—through governed receipts, explicit claim ceilings, and preserved terminal states—so that null, failure, and blocked outcomes remain part of model state rather than disappearing from the public record.

HydraDG demonstrates bounded custody and systems-validation behavior under that contract: failures are first-class, hashes are mandatory, and claim ceilings block over-interpretation. HydraLamp perturbation, tamper, concurrency, and replay results evaluate custody mechanics only and do not constitute treatment-effect evidence for EXP-008/009. Vithia appears in the companion framework lineage as a bounded exemplar implementation; it is not promoted to primary evidence for the reported experiments. The EXP-008/009 underpowered terminations bound what cannot yet be claimed while preserving malformed and timeout cells for audit.

6.1 Limitations

Limitations include bounded case replication, local-only model lanes in the reported matrix, and absence of runtime-equivalent successor replication in primary results. A preregistered Qwen 3.8 successor lane exists but is non-terminal and omitted from primary results. Whole-project hierarchical atomization (SeedGraph v1a validation) ran for multiple days but was interrupted during final Parquet serialization without a BUILD_RECEIPT; partial node/edge artifacts are not readback-safe, so terminal whole-corpus atomization is not established in this submission. Biological, protein-structure, and cellular perturbation applications of the custody framework remain future validation targets, not demonstrated generalizations of the agent-context results here.

6.2 Future directions

Agent systems. Larger successor replications, governed multi-agent handoffs, selective replay, and resource-aware custody gates under preregistered manifests.

Structured retrieval at scale. Checkpointed ingest and transactionally durable atomization so long hierarchy builds can resume from batch boundaries rather than materializing the entire graph at finalization.

Cross-project federation. Project-local FCG roots with explicit cross-project references and no automatic evidence admission between repositories.

Cryptographic custody. Authorized signatures and MMR commitments where policy permits; hashes remain identity commitments unless a verified signing operation occurred.

All directions above are *planned* unless independently measured; they do not extend the EXP-008/009 empirical claims.

7 Conclusion

We proposed the term *Mechanical Scientific Model* for a computational model class whose governed state transitions expose evidence, separate deterministic from probabilistic outputs, preserve failure-complete terminal states, and bind explicit claim ceilings with forward and reverse traceability. FCO/FCG provide the custody substrate; HydraDG is one implementation examined here under that proposal. Two preregistered HydraDG studies (EXP-008, EXP-009) closed as *underpowered*, demonstrating bounded custody and systems behavior while withholding confirmatory promotion of structured retrieval effects on failure-prevention endpoints. We recommend explicit Mechanical Scientific Method discipline—custody receipts, claim ceilings, and terminal-state preservation—as baseline infrastructure for NewInML-style agent evaluations, without claiming that structured retrieval efficacy has been established.

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A Full EXP-008 / EXP-009 statistical audit

Table 3 records terminal verdicts with parse rates reverse-traced to frozen verdict JSON receipts. p -values remain *not informative* under the preregistered underpowered design.

Table 3: Terminal preregistered study audit (appendix).

Study	Verdict	Raw cells	Valid parse rate
EXP-008	UNDERPOWERED	300	0.907
EXP-009	UNDERPOWERED	300	0.883

B Stage-2 M0/M1/M2 baseline

Stage-2 IC Failure Learning execution (414 scored rows across two models) closed as FAILURE_LEARNING_BEHAVIOR_IMPROVEMENT_NOT_ESTABLISHED; descriptive only.

C HydraLamp systems-validation matrix

See Table 4 for custody-mechanics outcomes (not treatment-effect evidence).

Table 4: Systems-validation matrix (appendix).

Validation	Scope	Outcome
Perturbation matrix	100 cells	100/100 chain verification
Synthetic tamper suite	8 modes	8/8 detected
Concurrent execution	10 runs	PASS (10 unique run IDs)
Replay/restart recovery	44 events	PASS
Live provider ladder	R0–R6	Bounded external failure preserved

D Deterministic verification and tooling

Figure 2 and DETERMINISTIC_AUDIT_TOOLING_LEDGER.jsonl document the custody verification pipeline with frozen script SHA-256 and R1/R2/R3 roots.

Source bytes → SHA-256 → Atom → ResultAtom → Claim/Table → FCG delta → exact-SHA verification

Figure 2: FIG-001: governed custody pipeline from source bytes to exact-SHA verification (structured data; self-generated).

E Citation and reference custody

Portfolio recovery bounded 185 occurrences to unique identities; only proposition-backed references are admitted. Citation fabrication literature informs desk-rejection risk [16], [17], [13], [18], [12].

F Requirement, template, and deadline drift

The NewInML workshop submission form requests a CC BY 4.0 article license for this non-archival track; OpenReview platform terms govern platform use separately from article and repository licenses. Repository code remains Apache-2.0 (separate layer).

G Prior-art comparison matrix

Related-work boundaries include PROV-O, CWLProv, and Workflow Run RO-Crate [16], [17], [13], [18], [12]. Prior-art search responses remain *DISCOVERY_ONLY*; not promoted to verified empirical novelty proof.

H SeedGraph / FCO / FCG custody audit

TOTAL_VERIFIED_INGEST_COMPLETE=NO with verified coverage 31.55% (307/973).

I Planned and nonterminal lanes

GPU SGLang canary, Q38 XENV, and full verified ingest remain nonterminal. All appendix citations resolve through the single main-text bibliography authority.

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